A square-root tangent removes the compound-Poisson branch

29 October 2026

The five-count argument already controls the full signed discrepancy at moderate noise. Its lower endpoint can be reduced from .075 to .03 without a new probability estimate. The only change is to use a better affine upper bound for $\sqrt{\lambda}$ in the curvature error. Four rational endpoint inequalities then replace the 291-cell negative-phase certificate in the selected $n \ge 200$ right-strip proof. The older $n \ge 42$ theorem retains its larger index domain and is preserved.

Theorem 1. *Let $n \ge 200$ be an integer, $p \in [2/5, 53/100]$, $q = 1-p$ and $v = pq$. Let $Y = B + U$, where B is Bernoulli of parameter p and $\mathbb{E}(U \mid B) = 0$. Write U_j for the conditional noise given $B = j$, and assume*

$$-3/5 \le U_0 \le a_0, \qquad -a_1 \le U_1 \le 3/5, \qquad a_0, a_1 \ge 0, \qquad a_0 + a_1 \le 9/125.$$

Set $s = \mathbb{E}U^2$, $\lambda = ns$, $X = (Y-p)/\sqrt{v+s}$ and $\rho = \mathbb{E}|X|^3$. If $3/100 \le \lambda \le 9/25$, then independent copies $X_1, \dots, X_n$ satisfy, for every raw threshold $|H| \le 46/25$,

$$\mathcal{E} := \sqrt{n}\{\mathbb{P}(X_1 + \cdots + X_n \le H) - \Phi(H/\sqrt{n})\} - C_E\rho < -\lambda/100, \qquad C_E = \frac{3+\sqrt{10}}{6\sqrt{2\pi}}. \quad (1)$$

The conclusion covers both signs of H, all phases and all threshold atoms, but concerns the positive signed discrepancy. Acquiring the strip for a maximizing witness, and applying it to a reflected law, are separate steps. This is not an unrestricted cutoff theorem.

1 The retained probability estimates

The relevant ancestors are the v13 weighted crossing identity, the v14 summable-curvature lemma, and the v15 five-count reserve. Their precise source locations are listed below. We give all estimates needed to join them. Write $t = np + w$, where $w = H\sqrt{v+s}$, and $k = \lfloor t \rfloor$, $u = t - k \in [0,1)$. Then $|w| < 231/250$ and the crossing identity reads

$$\mathbb{P}\Big(\sum X_i \le H\Big) = (1-u)F_B(k) + uF_B(k+1) + \mathcal{C}(t) - \sum_{i=1}^{n} \pi_i g_i(t), \quad (2)$$

where π_i and F_B are binomial masses and CDF values, and $g_i \ge 0$. The crossing is interior since $n(p-a_0) > |w|$ and $n(q-a_1) > |w|$.

For the five indices $k-1, \dots, k+3$, one has $|i - np| < 4$. Finite Stirling on the segment $[.38, .55]$, whose variance product exceeds .23, gives

$$\sqrt{n}\pi_i > \frac{199}{250}\left(1 - \frac{8 + 1/12}{200(23/100)}\right) > \frac{13}{20}.$$

The five-count reserve is obtained by putting $(2/5)\sin^2(\pi x)$, restricted to $[0,1]$, below the crossing kernel. Four centered replacements freeze the conditional sum at the middle count. Their total

cost is at most $10\pi^2\lambda/n$ before the factor $2/5$. The complete omitted-translate tail is bounded by the fourth moment divided by 16. With $\alpha = 39/40$, $b = 203/200$, $z = 3\alpha$ and $x = z\lambda$, this gives

$$\sqrt{n}\sum_i \pi_i g_i \ge W(\lambda) := \frac{13}{50}\left\{\frac{x - x^2/2 + x^3/6 - x^4/24}{2} - \frac{(9/25)b\lambda + 3b^2\lambda^2}{16} - \frac{10(22/7)^2}{200}\lambda\right\}. \tag{3}$$

Here the frozen variance lies between $(1 - 5/n)\lambda$ and $(1 + 3/n)\lambda$. Independence and the support bound give its fourth moment at most $(9/25)\tau + 3\tau^2$, and its characteristic function at 2π has modulus at most $e^{-3\tau}$. Finally $1 - e^{-x} \ge x - x^2/2 + x^3/6 - x^4/24$ for every $x \ge 0$. None of these estimates needs $\lambda \ge .075$.

For the summable curvature put

$$c = (np - q - t)_+, \qquad T = \frac{100}{47}\lambda, \qquad M = \frac{9}{25}T + 3T^2.$$

On the increasing side of the binomial mass the actual conditional variance is at most $\lambda/q \le T$, and its fourth moment is at most M. The retained curvature lemma, with mode coefficient $41/100$, yields

$$\sqrt{n}\mathcal{C}(t) \le \frac{41/100}{nv^{3/2}}\left\{\frac{T}{4} + c\sqrt{T} + \frac{27M}{256}\left(\frac{5}{3} + \frac{5c}{4}\right)\right\}. \tag{4}$$

Here $c < 23/50$ if $H \le 0$, and $c = 0$ if $H \ge 0$. The bound on the binomial mode and the hypotheses $c \le 1$ and $v^{3/2} > 117/1000$ are unchanged from the v15 derivation. All count tails in this lemma are summed analytically; no finite truncation is being used.

2 The tangent and the signed normal errors

The useful replacement is simply

$$\sqrt{\lambda} \le \frac{5}{3}\lambda + \frac{3}{20}, \qquad \sqrt{T} < \frac{3}{2}\sqrt{\lambda} \le \frac{5}{2}\lambda + \frac{9}{40}. \tag{5}$$

The first inequality is equivalent to $(\sqrt{\lambda} - 3/10)^2 \ge 0$; the second uses $100/47 < 9/4$. Define the quadratic polynomial

$$S_*(\lambda, c) = \frac{T}{4} + c\left(\frac{5}{2}\lambda + \frac{9}{40}\right) + \frac{27M}{256}\left(\frac{5}{3} + \frac{5c}{4}\right).$$

All its coefficients are nonnegative for the two choices of c used below.

Schulz's strict binomial theorem [1, Theorem 1] bounds both integer anchors in (2) by $C_E\rho_B$, where $\rho_B = (1 - 2v)/\sqrt{v}$. Convex interpolation preserves the strict bound. For $G(y) = \Phi((y - np)/\sqrt{nv})$, the signed interpolation remainder is at most

$$\sqrt{n}\{(1 - u)G(k) + uG(k + 1) - G(t)\} \le \frac{\phi(0)}{2nv^{3/2}}u(1 - u)(u - w)_+.$$

When $H \le 0$ the variance change from v to $v + s$ is favorable. Using $u(1 - u) \le 1/4$ and $u^2(1 - u) \le 4/27$, the total normal cost is at most

$$\frac{2/5}{nv^{3/2}}\left(\frac{231}{2000} + \frac{2}{27}\right).$$

When $H \ge 0$, its bound is instead

$$\frac{2/5}{nv^{3/2}}\left(\frac{2}{27} + \frac{231}{500}\lambda\right).$$

The second term comes from $s = \lambda/n$; it has no missing normalization. The fixed signs on the two conditional supports give

$$(v+s)^{3/2}(\rho - \rho_B) \geq s\{87/250 - (13/16)s\} > 0,$$

since $s \leq 9/5000$. Thus the actual cubic penalty is favorable.

Combining these facts with (3)–(5) gives $\mathcal{E} < -W(\lambda) + E_{\pm}(\lambda)/n$, where

$$E_{-}(\lambda) = \frac{(41/100)S_*(\lambda, 23/50) + (2/5)(231/2000 + 2/27)}{117/1000},$$
$$E_{+}(\lambda) = \frac{(41/100)S_*(\lambda, 0) + (2/5)(2/27 + 231\lambda/500)}{117/1000}.$$

The entire improvement is in E_{-}; the positive branch is unchanged.

3 Four exact comparisons, with no parameter cells

Put $P_{\pm} = W - E_{\pm}/200 - \lambda/100$. Both $E_{\pm}$ are convex quadratics with positive coefficients, while

$$W''(\lambda) = -\frac{13}{50}\left\{\frac{z^2}{4}\left[(z\lambda - 1)^2 + 1\right] + \frac{3b^2}{8}\right\} < 0.$$

Hence $P_{\pm}$ are strictly concave on the real line. Exact rational arithmetic gives the following lower bounds:

	$\lambda = 3/100$	$\lambda = 9/25$
P_{-}	$> 461/10^6$	$> 885/10^6$
P_{+}	$> 4654/10^6$	$> 11247/10^6$.

Concavity proves positivity on the whole closed interval. Since $E_{\pm} > 0$ and $n \geq 200$, this proves Theorem 1. The companion checker reconstructs these polynomials using exact fractions and checks the displayed second-derivative identity coefficient by coefficient. No numerical optimizer, parameter mesh or compound-Poisson computation is used.

Corollary 2. *Under the same support, centering, probability and threshold hypotheses, for $n \geq 200$ and $0 < \lambda \leq 9/25$ one has $\mathcal{E} < -\lambda/100$. At $\lambda = 0$ one has $\mathcal{E} < 0$.*

Proof. For $0 < \lambda \leq 3/100$, use the v11 direct comparison at its shifted integer anchor and $C = C_E > 2/5$, followed by the strict binomial theorem. Its phase interval $[-34/125, 91/125]$ has length one, so an anchor always exists. For $\lambda \geq 3/100$, apply Theorem 1. The common endpoint is included in both domains. When $\lambda = 0$, both centered conditional noises vanish and the strict binomial theorem applies directly. □

Exact dependency changes and limitations

All paths below begin with `mathematics/`.

- Retain `simplification-v15/2026-10-28/loss-direct.tex`, the five-count theorem and its proof; `simplification-v13/2026-10-28/crossing.tex`, the weighted identity; and `simplification-v14/2026-10-28/geometry-slack.tex`, the summable-curvature lemma. The finite-Stirling and signed normal estimates are the displayed ones in `simplification-v15/2026-10-28/right-direct.tex`.
- Retain `simplification-v11/2026-10-28/counts-v11.tex`, its first theorem, for positive noise through .03; its scalar Poisson-tail argument remains a dependency. Retain the external binomial theorem.

- In the $n \geq 200$ right-strip union of `simplification-v15/2026-10-28/join-review.md`, replace the .03–.075 phase split by Theorem 1. Neither `simplification-v12/2026-10-28/compound-proof.tex` nor the separate `simplification-v12/2026-10-28/outer.tex` is then needed on that path. Thus the 291-cell allocation certificate, compound-Poisson evaluator and its complete-tail and box-coverage subarguments leave this dependency path.

This does not replace the $n \geq 42$ anchor-comparison theorem at its original scope. In particular no statement for $42 \leq n < 200$ is inferred. The acquired-witness $n \geq 700$ union uses only $n \geq 200$ on its right side, so this replacement suffices there. Left-strip estimates, coarse Fourier entry, support acquisition, reflection, attainment and the genuine finite base are unchanged. No existing source or certificate is removed.

The old tangent $\sqrt{\lambda} \leq \lambda + 1/4$ gives $P_-(.03) < 0$ at $n = 200$; that is an obstruction to those sufficient bounds, not a counterexample to the probability theorem. The present tangent also cannot be extrapolated to zero noise: its scalar $P_-(0) < 0$. This explains why the v11 small-noise argument remains. Actual-law checks and these negative controls are recorded separately from continuous-domain acceptance.

References

[1] J. Schulz, *The optimal Berry–Esseen constant in the binomial case*, doctoral dissertation, Universität Trier, 2016. `https://d-nb.info/1197702695/34`.